

# Farhad Abedinzadeh

Postgraduate Researcher (PhD) in Bioengineering | Durham University

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## Research profile

I am a postgraduate researcher (PhD) in Bioengineering at Durham University working on interpretable AI for early prediction of childhood myopia using longitudinal multimodal retinal imaging and biometric data. My research interests include OCT/OCTA analysis, retinal biomarkers, ocular biometry, longitudinal modelling and multimodal learning.

My previous research spans biomedical image and signal processing, computational neuroscience and machine learning, including EEG connectivity, EMG, gait, voice and MRI analysis in neurodevelopmental and neurodegenerative disorders.

## Education

Current

### PhD in Bioengineering · Durham University, United Kingdom

Interpretable AI for early prediction of childhood myopia using longitudinal multimodal retinal imaging and biometric data.

2021-2023

### MSc in Biomedical Engineering (Sports Engineering)

Islamic Azad University, Mashhad Branch, Iran · GPA: 17.40/20

Thesis: Detection of ADHD Using Phase-Based Brain Connectivity and Graph Theory.

2015-2020

### BSc in Biomedical Engineering

Sadjad University of Technology, Mashhad, Iran

Thesis: Brain Tumor Detection Using Image Processing and Artificial Neural Networks.

## Teaching & research experience

Durham

### AI and Deep Learning · Python teaching materials

Preparation of practical notebooks and learning materials for students with limited Python experience.

May 2024

### Instructor · EEG Signal Processing Workshop

Workshop held by the National Brain Mapping Laboratory in Mashhad, Iran.

Jul-Oct 2022

### Instructor · Biomedical Image & Signal Processing Using MATLAB

Islamic Azad University, Mashhad Branch. Designed lectures, supervised practical sessions and guided projects in signal analysis, wavelets, pattern recognition and machine learning.

Joined Oct 2021

### Research Assistant & Lab Manager

Research with Seyyed Abed Hosseini, Islamic Azad University, Mashhad Branch. Work on brain connectivity and multimodal data in neurological disorders.

Started Jan 2022

### Biomedical programming & research project support

Python and MATLAB applications, data analysis, research methodology and support for undergraduate and master's research projects.

Joined Feb 2024

## Venture Capital Analyst of Medical Devices

Razavi HighTech Industries, Mashhad, Iran.

## Technical skills

### Programming & machine learning

Python, MATLAB, PyTorch, TensorFlow/Keras, scikit-learn and Git.

### Biomedical analysis

Image and signal preprocessing; EEG, EMG, ECG, gait and MRI analysis; connectivity, graph theory and time-frequency methods.

### Retinal imaging research

OCT/OCTA exploration, EyePy workflows, retinal segmentation, thickness maps, en face projections and biometric integration.

### Research tools

MNE, EEGLAB, Brainstorm, FieldTrip, BrainNet Viewer, BRAPH, HERMES, SPM, LaTeX, Mendeley, EndNote and SPSS.

## Selected publications

### Enhancing Multi-class Diabetic Retinopathy Detection Using Tuned Hyper-parameters and Modified Deep Transfer Learning

Medical imaging Multimedia Tools and Applications · 2024

Yeganeh Modaresnia, Farhad Abedinzadeh , Seyyed Abed Hosseini.

### Improved ADHD Diagnosis Using EEG Connectivity and Deep Learning through Combining Pearson Correlation Coefficient and Phase-Locking Value

EEG & connectivity Neuroinformatics · 2024

Elham Ahmadi Moghadam, Farhad Abedinzadeh , Seyyed Abed Hosseini, Mohammad Hossein Moattar.

### Simultaneous Time-Frequency Analysis of Gait Signals of Both Legs in Classifying Neurodegenerative Diseases

Gait & movement Gait & Posture · 2024

Farhad Abedinzadeh, Elham Ahmadi Moghadam, Seyyed Abed Hosseini.

### Hybrid Deep Transfer Learning Based Early Diagnosis of Autism Spectrum Disorder Using Scalogram Representation of Electroencephalography Signals

EEG & connectivity Medical & Biological Engineering & Computing · 2024

Farhad Abedinzadeh , Yeganeh Modaresnia, Mohammad Hossein Moattar.

Complete bibliography: <https://farhadabedinzadeh.github.io/publications/>

## Academic service

Peer reviewer for Biomedical Signal Processing and Control; Pattern Recognition Letters; Computers in Biology and Medicine; Physical and Engineering Sciences in Medicine; Multimedia Tools and Applications; BMC Public Health; and Iran Journal of Computer Science.

## Recognition

- Ranked second among Biomedical Engineering master's students at Islamic Azad University, Mashhad Branch.
- Finalist among 100 teams in the One Thousand Technological Ideas Tournament with an intelligent ring concept for HRV-based epileptic seizure prediction.

## Languages

Persian: native · English: advanced (C1, IELTS Academic).